

week-4(Pravalika-2520030402)

Question:

Write a C program where a parent process creates multiple child processes and synchronizes their completion using `wait()` and `waitpid()`. Compare both functions.

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t p1, p2, p3;

    p1 = fork();

    if (p1 == 0)
    {
        printf("Child 1: PID = %d\n", getpid());
        sleep(3);
        exit(1);
    }

    p2 = fork();

    if (p2 == 0)
    {
        printf("Child 2: PID = %d\n", getpid());
        sleep(2);
        exit(2);
    }

    p3 = fork();
```

```

if (p3 == 0)
{
    printf("Child 3: PID = %d\n", getpid());
    sleep(1);
    exit(3);
}

wait(NULL);
printf("wait() completed.\n");

waitpid(p1, NULL, 0);
printf("waitpid() for Child 1 completed.\n");

waitpid(p2, NULL, 0);
printf("waitpid() for Child 2 completed.\n");

waitpid(p3, NULL, 0);
printf("waitpid() for Child 3 completed.\n");

return 0;
}

```

Output:

```

rikki@BOOK-GJU4MP9458:~$ nano practical4.c
rikki@BOOK-GJU4MP9458:~$ gcc practical4.c
rikki@BOOK-GJU4MP9458:~$ ./a.out
Child 1: PID = 756
Child 2: PID = 757
Child 3: PID = 758
wait() completed.
waitpid() for Child 1 completed.
waitpid() for Child 2 completed.
waitpid() for Child 3 completed.
rikki@BOOK-GJU4MP9458:~$ |

```